

HECTOR GENARO PACHECO

Software Engineer | Full-Stack • Cloud • AI • IoT • Embedded Systems

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PROFESSIONAL SUMMARY

Software Engineer with a physical-world engineering background spanning full-stack development, AWS cloud infrastructure, retrieval-augmented AI systems, IoT edge computing, and industrial instrumentation & controls. Builds auditable software across cloud SaaS, native Android applications, GraphQL business platforms, governed RAG pipelines, and Raspberry Pi hardware-integrated systems.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++, Kotlin

Frontend & Mobile: React, Next.js, Android (Kotlin), Jetpack Compose, HTML5, CSS3

Backend & APIs: Node.js, Express, GraphQL, REST APIs, FastAPI, WebSockets, Zod

Embedded & Industrial: Raspberry Pi 5, GPIO, GPS/NMEA, Sensors, 4–20 mA Loops, PLC I/O

Cloud & Infra: AWS ECS Fargate, ECR, RDS, Secrets Mgr, KMS, VPC, Docker, Vercel, Linux, Git

Data: PostgreSQL, Prisma, MongoDB, Supabase, Apollo, pgvector

AI / Retrieval: OpenAI API, RAG, Embeddings, pgvector, Semantic Retrieval, Structured LLM

AI & Data Platforms: Palantir Foundry, Palantir AIP, AI Workflow Training, Data Governance

FEATURED DEPLOYED & ENGINEERING PROJECTS

CalTrack — *Industrial Calibration SaaS & Mobile* [React](#) • [TypeScript](#) • [Node](#) • [Express](#) • [PostgreSQL](#) • [AWS ECS](#) • [Kotlin](#) / [Android](#) (Jetpack Compose)

[Platform](#) [LIVE APP](#) [GITHUB](#)

- Architected and deployed a full-stack industrial calibration platform for instrument tags, calibration records, RBAC, and audit histories.
- Containerized Node/Express backend with Docker on AWS ECS Fargate, ECR, RDS PostgreSQL, Secrets Manager, KMS, and Vercel.
- Engineered a native Android companion application in Kotlin using Jetpack Compose, Material 3, CameraX, and ML Kit for QR/barcode scanning.

Procurement Tracking System — *Enterprise Procurement Workflow Platform* [React](#) • [TypeScript](#) • [GraphQL](#) • [Apollo](#) • [Express](#) • [Prisma](#) • [PostgreSQL](#)

[LIVE APP](#) [GITHUB](#)

- Built a deployed procurement operations platform supporting purchase requests, approval workflows, vendor management, purchase orders, goods receiving, and multi-role RBAC.
- Implemented GraphQL APIs with Apollo Server/Client, Prisma/PostgreSQL persistence, JWT auth, and resolver permission enforcement.

Thriveward Funding Intelligence — *Retrieval-Augmented Funding Intelligence Platform* [TypeScript](#) • [Express](#) • [PostgreSQL](#) • [pgvector](#) • [OpenAI](#) • [RAG](#)

- Built a governed RAG pipeline spanning PDF document ingestion, page-bounded chunking, OpenAI embeddings, pgvector cosine retrieval, grounded evidence construction, and structured gpt-4o-mini analysis.
- Engineered index-integrity controls using provider/model/configuration hashes and versioned indexes to prevent stale or incompatible embeddings from silently contaminating retrieval results.

FieldTrack AI — *Edge Computing & Sensor Telemetry Platform* [GITHUB](#)

[Raspberry Pi 5](#) • [Python](#) • [FastAPI](#) • [Node](#) • [React](#) • [WebSockets](#)

- Built an edge-computing platform combining Python hardware services, GPS/NMEA parsing, PIR motion sensing, camera/computer-vision processing, and real-time React telemetry over WebSockets.

PRODUCTION CLIENT WORK & ADDITIONAL PROJECTS

OC Water Features — *Production Full-Stack Web Platform* [LIVE SITE](#)

[Next.js](#) • [TypeScript](#) • [Supabase](#) [PostgreSQL](#) • [Zod](#) • [Vercel](#)

- Built and deployed a production web platform with SEO, lead intake workflows, customer-facing forms, administrative tooling, and persistent database-backed workflows.

Additional Engineering Repositories: [Pi Arcade OS](#) — Embedded Hardware & Software Gaming Platform (Raspberry Pi / Python / Pygame / GPIO)

[PROJECT SITE](#) [GITHUB](#) • [E-Commerce Store](#) — Full-Stack GraphQL Store (React / GraphQL / MongoDB / AWS S3) [GITHUB](#)

TECHNICAL & INDUSTRIAL EXPERIENCE

Instrumentation & Controls Technician — *Process Control & Industrial Systems*

Field Engineering Experience

- Calibrated, commissioned, and troubleshot 4–20 mA process transmitters (pressure, temperature, flow, level), PLC I/O wiring harnesses, terminal blocks, relays, and field instruments across 480VAC, 120VAC, and 24VDC power systems.
- Verified process control loop operation, signal integrity, and safety shutdowns while maintaining auditable calibration logs.

EDUCATION & PROFESSIONAL TRAINING

Western Governors University (WGU) — B.S. Software Engineering (In Progress, 2026–Present)

Palantir Technologies — Foundry & AIP Training (May 2026)

Mirion Technologies — SU-890 iCAM Operations & Maintenance (March 2026, 16.00 Continued Education Credits)